

WEST BEND, WI, USA

WMO: 726413

Lat: 43.417N

Lon: 88.133W

Elev: 269

StdP: 98.13

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04875

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-17.6	-26.2	0.4	-19.7	-22.9	0.5	-16.6	10.5	-6.5	9.3	-6.6	3.2	280	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	31.2	22.5	29.1	21.4	27.8	20.6	23.9	28.4	22.9	27.7	22.1	26.5	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.7	26.8	21.4	16.6	25.4	20.9	16.1	25.0	72.9	28.4	69.3	27.5	66.0	26.4	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.5	DB	-23.7	33.5	2.8	1.8	-25.7	34.8	-27.3	35.9	-28.9	36.9	-30.9	38.2	(4)
(5)			WB	-24.0	25.5	2.7	1.2	-26.0	26.4	-27.6	27.1	-29.1	27.8	-31.1	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-6.2	-5.1	0.9	7.1	13.2	18.6	21.2	20.1	16.4	9.7	3.1	-3.1	(6)	
	DBStd	10.84	6.55	6.15	6.14	5.05	4.75	3.83	3.32	3.02	4.26	4.83	5.23	5.85	(7)	
	HDD10.0	2046	503	424	291	117	21	0	0	0	3	66	213	407	(8)	
	HDD18.3	4016	762	657	542	339	171	42	10	16	87	271	456	664	(9)	
	CDD10.0	1339	0	0	8	29	121	259	348	314	195	57	7	1	(10)	
	CDD18.3	268	0	0	1	2	12	51	100	71	28	3	0	0	(11)	
	CDH23.3	2383	0	0	4	33	140	482	910	549	234	31	0	0	(12)	
	CDH26.7	627	0	0	0	6	28	138	272	126	52	4	0	0	(13)	
(14) Wind	WSAvg	3.0	3.6	3.5	3.3	3.7	3.1	2.6	2.3	2.1	2.4	3.1	3.3	3.4	(14)	
(15) Precipitation	PrecAvg	813	31	25	50	74	74	89	97	93	106	69	61	41	(15)	
	PrecMax	1052	96	78	140	115	166	184	262	189	267	146	172	98	(16)	
	PrecMin	575	2	2	8	24	10	36	27	10	6	5	5	0	(17)	
	PrecStd	130	20	18	31	25	36	44	49	44	71	38	38	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	12.7	22.5	27.3	29.8	32.7	34.1	32.2	30.9	27.1	19.6	13.9	(19)	
		MCWB	7.7	8.3	15.4	16.9	20.4	22.7	24.0	23.2	21.6	20.0	13.8	13.2	(20)	
	2%	DB	7.1	8.0	17.1	22.8	27.2	30.2	31.3	29.8	27.9	22.9	16.3	8.9	(21)	
		MCWB	4.8	5.2	11.9	14.9	18.5	21.4	23.1	21.8	20.5	17.4	12.3	7.5	(22)	
	5%	DB	3.9	5.1	12.7	19.2	24.7	27.8	29.2	27.7	26.2	21.0	13.6	7.0	(23)	
		MCWB	2.5	2.9	8.4	12.7	17.1	20.3	21.8	20.9	19.5	16.2	10.3	5.3	(24)	
	10%	DB	2.3	2.6	9.9	16.2	22.2	26.2	27.7	26.4	23.9	17.6	11.3	4.1	(25)	
		MCWB	1.5	1.1	6.4	10.2	15.6	19.1	21.0	19.9	18.5	13.8	8.5	2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	9.0	16.1	19.6	21.7	24.8	25.7	24.7	23.4	21.6	15.6	12.9	(27)	
		MCDB	8.7	12.1	21.4	23.4	27.6	29.9	31.0	29.4	27.9	25.4	17.5	14.1	(28)	
	2%	WB	4.6	5.8	13.0	15.7	19.9	22.9	24.1	23.4	22.0	19.3	13.1	7.8	(29)	
		MCDB	6.5	7.8	16.8	21.8	25.0	28.5	28.6	27.6	26.1	21.8	15.8	8.4	(30)	
	5%	WB	2.8	3.4	9.7	13.6	18.4	21.7	23.1	22.4	20.9	16.8	10.9	5.4	(31)	
		MCDB	4.0	5.1	12.7	18.7	23.0	26.7	27.9	26.3	24.1	19.5	13.0	6.7	(32)	
	10%	WB	1.3	1.3	6.4	10.8	16.8	20.3	22.3	21.3	19.7	14.4	8.4	3.0	(33)	
		MCDB	2.2	2.6	9.6	15.6	20.9	24.5	26.8	24.6	23.3	17.3	10.9	4.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.2	8.9	10.3	11.1	11.0	11.1	10.7	11.3	9.6	8.1	6.7	(35)	
		MCDBR	8.7	10.3	13.0	16.0	14.5	12.8	12.1	12.1	13.1	13.6	11.9	8.7	(36)	
	5% WB	MCWBR	7.3	7.8	8.7	9.2	7.9	6.4	5.6	5.9	6.7	8.0	8.4	7.4	(37)	
		MCWBR	8.3	9.7	12.2	14.7	13.2	11.1	10.6	10.3	11.0	11.3	10.7	8.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.280	0.315	0.338	0.385	0.405	0.424	0.416	0.408	0.378	0.340	0.302	0.281	(40)	
		taud	2.394	2.284	2.360	2.271	2.262	2.255	2.277	2.292	2.363	2.468	2.509	2.443	(41)	
	Ebn at Noon		857	875	901	879	868	850	852	847	847	842	829	824	(42)	
		Edh at Noon	76	102	108	128	133	134	130	124	107	85	69	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.53	2.53	3.67	4.56	5.37	6.01	6.14	5.26	4.26	2.71	1.71	1.21	(44)	
	RadStd		0.21	0.28	0.39	0.42	0.44	0.42	0.28	0.33	0.35	0.31	0.14	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page